



Modern Talking...

Is the time ripe for Skype, or is it all just hype?

P2P networks took off with Napster, and you've seen a lot of them, as well as P2P clients, over the past few years—the Gnutella network, eDonkey, KaZaA, Shareaza, Morpheus, to name a few.

However, VoIP (Voice over IP) never really got off to a start. The reasons cited for its failure range from poor sound quality, to calls not lasting long enough. From systems unable to get calls past firewalls to the big NAT problem. Dubious as it may sound, Network Address Translation (NAT) renders over a whopping 50 per cent of home PCs unable to communicate with older VoIP software.

Then there is Skype, a brainchild of the people behind KaZaA—Janus Friis and Niklas Zennstrom. It's VoIP over P2P now. The revenue reduction that has jolted the recording industry may have the same impact on telecom companies, if everyone with a just a computer and a modem began talking to each other!

"Best of all, Skype does not require you to reconfigure your firewall or router—it just works!", boasts the Skype Web site. Imagine the magnitude of repercussions. As you are reading this, you could be downloading Skype for free, and within a minute, be talking to anyone else who installed Skype, anywhere in the world! And Skype even promises that voice quality is "better than your regular telephone." So the bells are tolling for the telecom industry, right?

That is, admittedly, too drastic a conclusion, since there are several other things that need to be weighed. However, three things are for sure: P2P is here to stay; Friis and Zennstrom's new idea may be more revolutionary than their first; and, of course, the telcos aren't breathing easy since Skype entered the scene.

So how exactly does Skype work? It works on the same principle as KaZaA, and similar clients do. No single server

maintains a database, or is responsible for the collection and broadcasting of data. It's all distributed. Your voice data gets carried to a computer close to you—a neighbour. Immediately, this machine finds another neighbour closer to your destination and transmits the data there. The relay carries on until your data packet gets to the Skype user you are calling.

Actually, it is not as simple as that, of course. There are issues to be tackled, such as Firewall and NAT traversal, hurdles to be dealt with. In this case, Skype uses 'The

Global Index' technology, which is a multi-tiered network. It works around supernodes, a special type of nodes that communicate in such a way that every node in the network has full knowledge of all available users and resources. So one major problem, that of a user knowing that another is online, is done away with. Within seconds of your logging on, the activity is propagated all over the network.

There are other concepts involved too, such as the fact that as more users are added to the network, the computational and bandwidth resources of the system increase proportionately.

Interestingly, while you fear privacy and security concerns, every packet on the Skype system is encrypted end-to-end. It effectively nullifies chances of eavesdropping.

Well, the story does not end with Skype. VoIP competitors are about to join in the race as well. Voipster, an Amsterdam-based company, is all set to launch a corporate competitor to Skype this month-end. Apart from targeting business users, the company will charge a monthly fee for unlimited calls, four-way conferencing, voicemail and sharing documents.

So, will Skype live up to its hype? Probably, yes. As of now, the Skype Web site reports close to 12 million downloads. In fact, Skype is the self-reported "fastest-spreading software on the Internet." Whether it will pass the test of time—only time will tell. 



WHINE LIST

How broad is your band?

Broadband: (*Adjective*) A data transmission technique allowing multiple high-speed signals to share the *bandwidth* of a single cable via frequency division multiplexing.

Bandwidth: (*Noun*) A data transmission rate; the maximum amount of information (bits per second) that can be transmitted along a channel.

Now you're probably wondering why *Digit* is telling you the meanings of these two words. Because no home-user in India has ever experienced this elusive thing called 'broadband'.

Cable operators and ISPs shamelessly display this prominently, on hoardings, on the flyers, everywhere. The truth of the matter is that not a single ISP, so far, is offering speeds that would rival what even dial-up users in the US take for granted. If the US is the bandwidth capital of the world, then we're a remote village on the outskirts of civilisation.

Does this mean there's no bandwidth in India? No, it means that there's no such thing as 'affordable bandwidth' here. Stop dreaming of streaming video, and consider yourself lucky to be able to download that 200 MB game demo in two days—there are people worse off you know, and all of them in a small town near you!

The only hope we have is that MTNL's flat rate Internet scheme forces other ISPs to give you your capped 'broadband' connection for even cheaper.

Welcome to bandwidth hell!